Matrix inequalities and norms

Minimum number of orthogonal conjugations for diagonal pinching

Difficulty: extreme **Importance:** interesting to the community

Problem statement

For each integer $d \geq 1$, define $\Delta(X) = \operatorname{diag}(x_{11}, \dots, x_{dd})$ for $X = (x_{ij}) \in \mathbb{R}^{d\times d}$. Determine the integer

$$\varphi(d) = \min\left\{q \geq 1 : \exists U_1, \dots, U_q \in O(d)\ \forall X \in \mathbb{R}^{d\times d}, \quad \Delta(X) = \frac{1}{q}\sum_{i=1}^{q} U_i X U_i^T\right\},$$

where $O(d) = \{U \in \mathbb{R}^{d\times d} : U^T U = I_d\}$. The same list of orthogonal matrices must work for every X. The minimum is finite: $q = 2^{\lceil \log_2 d\rceil}$ is always possible.

Why it matters

Diagonal extraction is a basic matrix operation. Expressing it by the shortest average of orthogonal similarities gives an exact decomposition complexity and sharpens real versions of majorization-based norm estimates.

References

1. J.-C. Bourin and E.-Y. Lee, *Averages over matrix unitary orbits and spectral order*, arXiv:2606.15624v2 (18 June 2026), Lemma 4.1 and Question 4.8. Primary text.
2. R. Bhatia, *Pinching, trimming, truncating, and averaging of matrices*, American Mathematical Monthly 107(7) (2000), 602–608, equation (2), for the complex unitary counterpart cited in reference 1. DOI.

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The latest source is v2 and explicitly asks for the smallest length. Searches included *orthogonal pinching minimum phi Bourin Lee*, *Averages over matrix unitary orbits spectral order 2026*, and *diagonal pinching orthogonal matrices minimum*. No general determination was located. The adjacent Question 4.9 reverses the quantifier order by allowing the matrices to depend on X; it is not counted separately here. This is an exact optimization question rather than a conjectured closed formula.